

How to **measure** anonymity in networks and create **algorithms** to **minimize** disclosure control risks?

Introduction

Analyzing social networks using methods such as centrality ranking or community detection, may offer unique insights into societally relevant problems such as segregation, cohesion and economic mobility. However, sharing network data can lead to identification risk of individuals represented.

Approach: k -anonymity

Counteract this risk with k -anonymity:

- Network is k -anonymous if all nodes are k -anonymous
- Node is **k -anonymous** if it is equivalent to $\geq k-1$ nodes
- For now: **unique** $\rightarrow$ **not anonymous**

Measure based on *attacker scenario*.

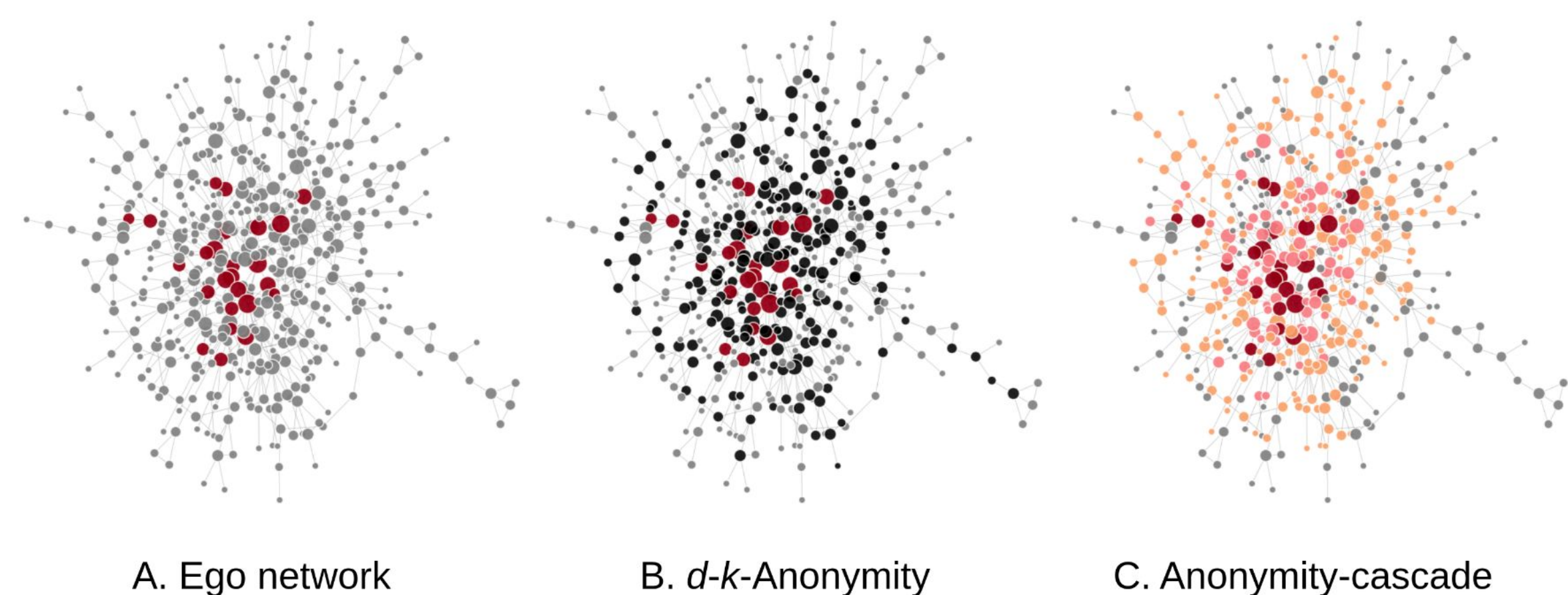
Approach: Uniqueness is infectious

d - k -Anonymity:

- Structure of 1- or 2-neighborhood (A, B)

Anonymity cascade:

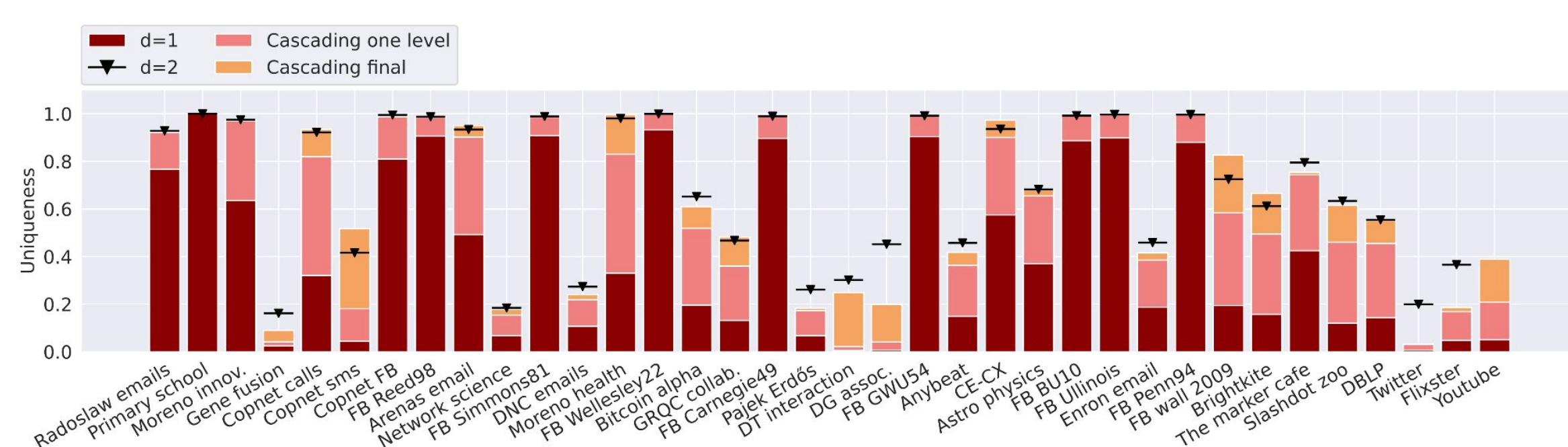
- Structure of 1-neighborhoods + existence of connection (C).
- "Infect" neighbors with uniqueness
- For one or multiple levels



Result: Anonymity in real-world networks

Even knowledge about 1-neighborhoods + connection is very de-anonymizing!

$\rightarrow$ **Need to look beyond the 1-neighborhood**



Based on: de Jong, Rachel G., Mark P.J. van der Loo, and Frank W. Takes. "The effect of distant connections on node anonymity in complex networks." *Scientific Reports* 14.1 (2024): 1156.

Stay in touch!

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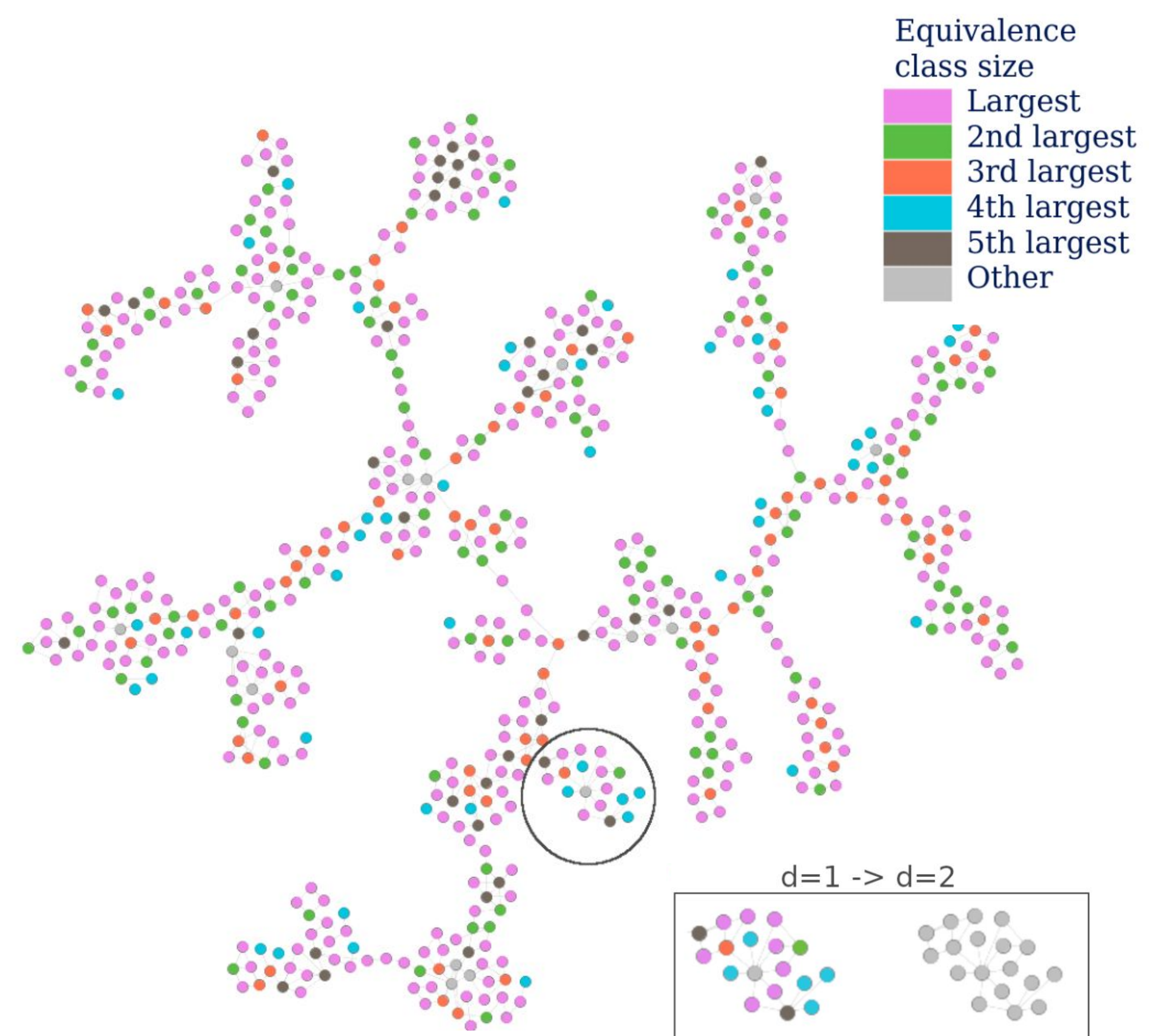
Publications



Example: family network

Part of **Dutch population network** parent-child relations.

- Same color $\rightarrow$ equivalent d -neighborhood



Fuzzy k -anonymity

It is likely that an attacker is not 100% certain about the information found. Additionally, data changes over time and may be incomplete, or overrepresenting certain populations.

How does accounting for uncertainty affect anonymity?

Approach: ϕ - k -anonymity

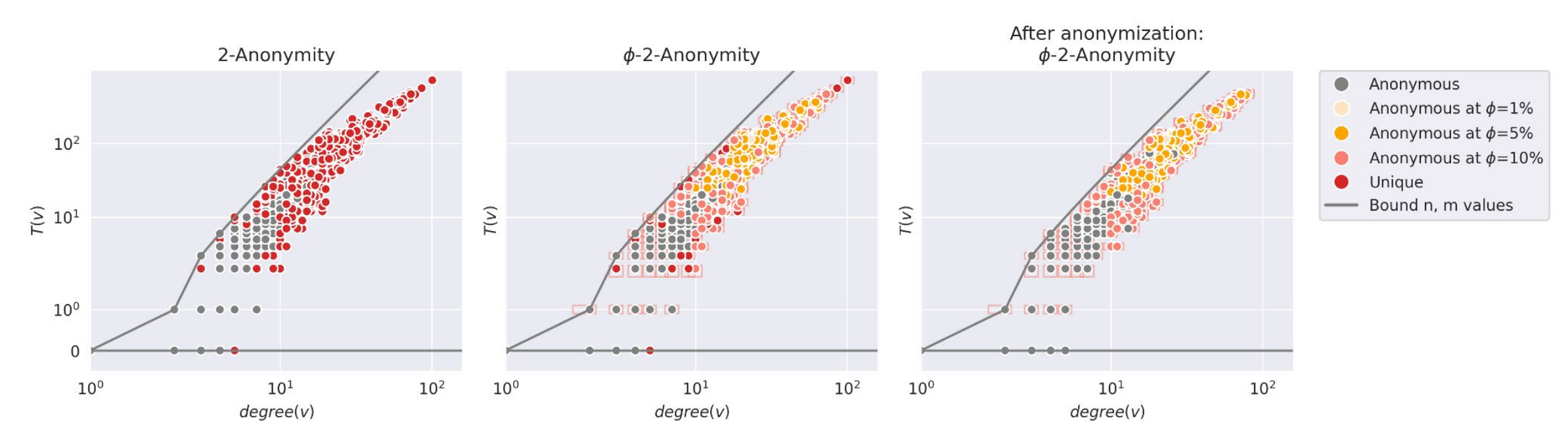
Nodes are equivalent if:

- Same degree, deg , part of same nr. triangles T

Nodes v, w are ϕ -similar if:

$$\bullet \frac{\Delta deg}{deg(v)} \leq \phi \quad \text{and} \quad \frac{\Delta T}{T(v)} \leq \phi$$

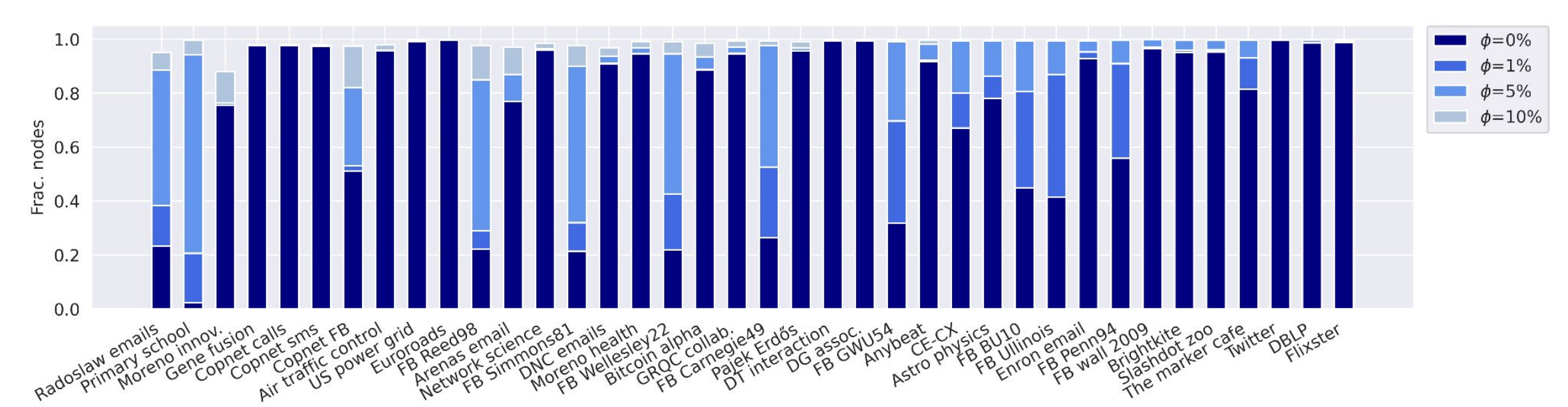
Example: Copnet facebook friends



Result: ϕ - k -anonymity in real-world networks

$\phi=0.05$: over 76% of nodes anonymous

$\rightarrow$ **Many unique nodes are quite similar to at least one other node**



Result: ϕ - k -anonymity + greedy anonymization

$\phi = 0$: limited effect anonymization

$\rightarrow$ **$\phi=0.10$ + Greedy: sufficient to anonymize these networks**

